

EXPERIENCE

Quantum Communication System Engineer

Terra Quantum
St. Gallen, Switzerland

August 2024 — Present

- **Led a team of three engineers** in developing a Key Management System (**KMS**) for a full-stack quantum key distribution (**QKD**) system.
- Drove **cross-functional collaboration** with hardware teams and product management to deliver **standards-compliant** quantum communication solutions adhering to **NIST** and **ETSI** guidelines with automated testing.
- Architected backend systems using **Python**, **PostgreSQL**, and **Celery** in **dockerized** environments.
- Implemented post-quantum cryptography (**PQC**) algorithms and certificate management solutions (CA/TLS).
- Built **Grafana** dashboards integrating real-time hardware health metrics via **Prometheus** and application logs via **Loki**, enabling comprehensive system observability.

Software Engineer

Restaurant Supply Chain → AdTech

Delivery Hero

Berlin, Germany

October 2022 — July 2024

- Developed and maintained **Flask** APIs (**Python**, **SQLAlchemy**) in line with CI/CD best practices.
- Integrated third-party APIs (**Shopify**, **Stream Chat**, **Twilio**) to expand software capabilities.
- Deployed and managed **AWS** services (**S3**, **Lambda**, **ECR**, **SQS**) with **Terraform**.
- Adapted **Dart**-based **Flutter** mobile app codebase to support recent upstream API changes.
- Improved performance of API gateways in **Go** resulting in **40%** latency reduction through custom caching middleware for **Go-chi** router. Accelerated **React** frontend loading times.
- Enabled automatic generation of **Swagger** API documentation for Go-chi services using swaggo.

Software Engineer

360dialog

Berlin, Germany

July 2020 — September 2022

- Worked with **WhatsApp Business Platform** API to ensure a seamless client messaging experience.
- Automated billing with **Stripe** API, significantly reducing the amount of manual work.
- Reduced client onboarding time from **days** to just **a few minutes** using Embedded Signup.
- Designed and implemented **Flask** APIs (**Python**, **SQLAlchemy**).
- Deployed and managed **dockerized** services on **GCP**.
- Implemented **behavior-driven development** (BDD) with tests in natural language using **behave**.
- Enhanced monitoring with **Grafana** and **Prometheus**, integrating alerts with **Slack**.

Quantum Software Engineer Intern

Wehner Group, Quantum Internet Division

QuTech

Delft, Netherlands

September — November 2019

- Developed embedded firmware in **C** for the Hercules LaunchPad microcontroller to control quantum physical setup via ADwin-Pro connection.
- Collaborated with physicists to build a **Reinforcement Learning** system in **Python** to control setup of lasers during the experiments with NV-center in diamonds.

Machine Learning Engineer

ChatFirst

Moscow, Russia

September 2018 — April 2019

- Led **NLP** initiatives: investigated, stacked, and utilized ML models in **Python** to improve chatbot performance.
- Enhanced the production system with **BERT**, fine-tuned on the company's dataset for downstream tasks such as chatbot knowledge retrieval.

EDUCATION

M.Sc. in Computer Science and Physics

Moscow Institute of Physics and Technology, Russia

2019 — 2021

B.Sc. in Computer Science and Physics

Moscow Institute of Physics and Technology, Russia

2014 — 2019

TECHNOLOGIES & LANGUAGES

- **Backend:** Python, Go, Java, C/C++
- **Web & Mobile:** Flask, Django, React, JavaScript, CSS/HTML, Flutter
- **Cloud & DevOps:** AWS/GCP, Docker, Terraform, Jenkins, Spinnaker
- **Monitoring:** Grafana, Prometheus, Datadog
- **Databases:** PostgreSQL, SQLAlchemy, peewee
- **ML:** PyTorch, TensorFlow, numpy, scikit-learn, pandas, NLTK, Gensim
- **English:** Full Professional, **German:** C1, **Russian:** Native